

I. THEOREM ON COMPLEX ANTISYMMETRIC MATRICES

For any complex antisymmetric matrix $A = B + iC$ where:

- $B^T = -B$ (real antisymmetric part)
- $C^T = -C$ (real antisymmetric part)

the following conditions hold:

1. A is normal ($AA^\dagger = A^\dagger A$) if and only if the commutator $[B, C] = BC - CB = 0$
2. When normal, A is unitarily diagonalizable with purely imaginary eigenvalues
3. When $[B, C] \neq 0$, A may have defective eigenvalues requiring Jordan form

Part 1: Normality Condition

For $A = B + iC$ with $B^T = -B$ and $C^T = -C$:

$$\begin{aligned} AA^\dagger &= (B + iC)(B^T - iC^T) \\ &= (B + iC)(-B - iC) \\ &= -B^2 - iBC - iCB + C^2 \end{aligned}$$

$$\begin{aligned} A^\dagger A &= (B^T - iC^T)(B + iC) \\ &= (-B - iC)(B + iC) \\ &= -B^2 - iCB - iBC + C^2 \end{aligned}$$

The normality condition reduces to:

$$AA^\dagger - A^\dagger A = -2i[B, C] = 0 \quad \Longleftrightarrow \quad [B, C] = 0 \quad (1)$$

Part 2: Diagonalizability

When $[B, C] = 0$:

- A is normal $\Rightarrow$ unitarily diagonalizable via $A = UDU^\dagger$
- Eigenvalues are purely imaginary since $A^\dagger = -A$

When $[B, C] \neq 0$:

- A is non-normal
- May have defective eigenvalues (geometric multiplicity $<$ algebraic multiplicity)
- Requires Jordan form $A = PJP^{-1}$ with non-trivial Jordan blocks